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KPK BOARD TEXTBOOK NOTES - MATHEMATICS CLASS 9

Unit 9: Introduction to Coordinate Geometry

Complete, easy English notes based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - the distance formula, the midpoint formula, collinear points, and recognising triangles and quadrilaterals from their coordinates, with worked examples and exam practice.

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> **How to use these notes:** Keep a calculator-free habit: most board questions use small whole numbers and the 3-4-5 triangle, so look for that pattern before you reach for long working. After each formula, cover the worked answer and try the example yourself - in this unit, method marks come from showing the substitution step.

What this unit covers

1. Points, axes and ordered pairs - a short revision of Unit 8
2. The distance formula and where it comes from
3. Using the distance formula, including distance from the origin
4. The midpoint formula for a line segment
5. Collinear points - three points on one straight line
6. Kinds of triangle proved by their side lengths
7. Quadrilaterals proved by sides, diagonals and midpoints
8. Formula summary, common mistakes, MCQs and practice questions with answers

Source book: KPTB Mathematics for Grade IX, Unit 9 (book pages 210-232). The notes follow the book's order and syllabus. The scan used for verification carries no printed copyright page, so no printing year is claimed here - the 17-unit structure is the current circulating KPTB edition.

1. From Points to Numbers - Revision

Coordinate geometry (also called analytic geometry) joins algebra and geometry: every point is described by numbers, so lengths and shapes can be worked out by calculation instead of by measuring a drawing.

- A point is an **ordered pair (x, y)**: x is the distance across from the origin, y is the distance up or down.
- The signs of x and y place the point in one of four **quadrants**: I (+, +), II (-, +), III (-, -), IV (+, -).
- A point on the x-axis has **y = 0**, like (5, 0). A point on the y-axis has **x = 0**, like (0, -3). The origin is O (0, 0).
- A **line segment** is the straight part joining two points, for example segment AB joining A and B. Its length and its middle point are the two calculations of this unit.

Why calculation beats measuring

A drawn diagram can be small, rough or impossible to draw to scale. Coordinates are exact: if two lengths calculated from coordinates are equal, the lengths really are equal, and a shape can be proved to be a square, a rectangle or a right-angled triangle without drawing it at all. That proving work is the heart of this unit and of the board questions on it.

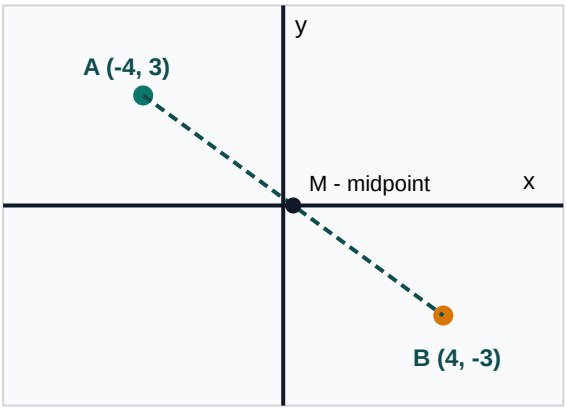


Figure 1. Segment AB joins A (-4, 3) and B (4, -3). This unit finds its length and its midpoint M by formula.

> **The two questions of this unit:** How long is the segment between two points? And where exactly is its middle? Two formulas answer both - learn them perfectly and the rest of the unit is applying them.

2. The Distance Between Two Points

Take two points A (x1, y1) and B (x2, y2). The horizontal gap between them is (x2 - x1) and the vertical gap is (y2 - y1). Those two gaps and the segment AB form a right-angled triangle, so **Pythagoras' theorem** gives the length of AB.

Distance formula: AB = sqrt($(x2 - x1)^2 + (y2 - y1)^2$)

Read it as: square the gap in x, square the gap in y, add, then take the square root.

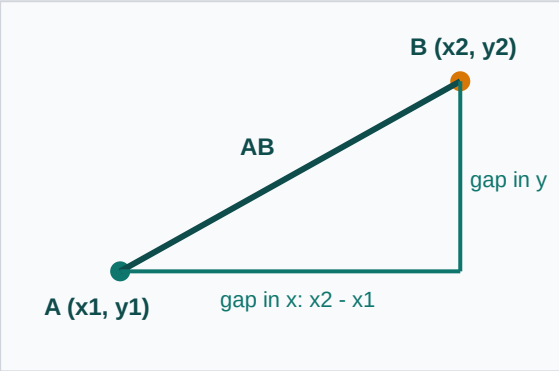


Figure 2. AB is the sloping side of a right-angled triangle whose other sides are the x-gap and the y-gap.

Example 1. Find the distance between A (2, 3) and B (5, 7).

Gap in x: 5 - 2 = 3. Gap in y: 7 - 3 = 4. AB = sqrt($3^2 + 4^2$) = sqrt(9 + 16) = sqrt(25) = **5 units**. Notice the 3-4-5 pattern - it appears again and again in board questions.

Example 2. Find the distance between P (-2, 3) and Q (4, -5). Gap in x: 4 - (-2) = 6. Gap in y: -5 - 3 = -8. PQ = sqrt($6^2 + (-8)^2$) = sqrt(36 + 64) = sqrt(100) = **10 units**. The minus sign disappears when the gap is squared, so a negative gap is never a problem.

3. Using the Distance Formula

Distance from the origin

The origin is just the point (0, 0), so the formula shortens. The distance of P (x, y) from the origin is:

$$OP = \sqrt{x^2 + y^2}$$

Example 3. Distance of (6, 8) from the origin: sqrt(36 + 64) = sqrt(100) = **10 units**. Distance of (-3, 4) from the origin: sqrt(9 + 16) = sqrt(25) = **5 units** - the signs make no difference after squaring.

Rules that keep the working safe

- **Either point can be first.** (x2 - x1) and (x1 - x2) differ only in sign, and the square removes the sign, so AB and BA give the same length.
- **Distance is never negative.** If your final answer is negative, a sign error has crept in - check the subtractions.
- **Keep the square root when it is not exact.** sqrt(13) or sqrt(18) = 3 sqrt(2) are proper answers. Do not round unless the question asks.
- **Same x or same y is easy.** If two points share an x value, the segment is vertical and its length is just the gap in y. Points (2, 1) and (2, 6) are 5 units apart. If they share a y value, use the gap in x.

Example 4. Find the distance between A (1, -2) and B (1, 4), and between C (-3, 5) and D (2, 5).

A and B share x = 1, so AB = 4 - (-2) = **6 units**. C and D share y = 5, so CD = 2 - (-3) = **5 units**. The full formula gives the same results: sqrt(0 + 36) = 6 and sqrt(25 + 0) = 5.

Exam trap: Subtract carefully with negative numbers. 4 - (-2) = 6, not 2. Write each gap on its own line first, exactly as in the examples, before squaring anything.

4. The Midpoint of a Line Segment

The **midpoint** is the point exactly halfway between two endpoints. Its x value is halfway between the two x values, and its y value is halfway between the two y values - in other words, the average of each pair.

Midpoint formula: M = ((x1 + x2) / 2 , (y1 + y2) / 2)

Add the x values and halve. Add the y values and halve. That is the whole method.

Example 5. Find the midpoint of A (2, 4) and B (6, 8).

x: (2 + 6) / 2 = 4. y: (4 + 8) / 2 = 6. Midpoint M = **(4, 6)**. Check with distances: AM = sqrt(4 + 4) = sqrt(8) and MB = sqrt(4 + 4) = sqrt(8) - equal, as a midpoint requires.

Example 6. Find the midpoint of P (-3, 5) and Q (5, -1).

x: (-3 + 5) / 2 = 1. y: (5 + (-1)) / 2 = 2. Midpoint M = **(1, 2)**. Averages work perfectly well with negative numbers - just add with care.

Finding a missing endpoint

If the midpoint and one endpoint are known, the other endpoint can be recovered: each coordinate of the missing endpoint is twice the midpoint coordinate minus the known endpoint coordinate.

Example 7. M (3, 1) is the midpoint of AB, and A is (1, 4). Find B. x of B: 2 x 3 - 1 = 5. y of B: 2 x 1 - 4 = -2. So **B = (5, -2)**. Check: ((1 + 5) / 2, (4 - 2) / 2) = (3, 1) = M. Correct.

Do not mix the formulas: Distance *subtracts* the coordinates and squares; midpoint *adds* the coordinates and halves. Writing the wrong one is the commonest slip in this unit.

5. Collinear Points

Points that lie on **one single straight line** are called **collinear** points. Three points A, B and C are collinear exactly when the two shorter distances add up to the longest distance.

Collinear test: AB + BC = AC (using whichever point lies in the middle)

The method - three distances, one addition

1. Find all three distances: AB, BC and AC.
2. Add the two smaller ones.
3. If the sum equals the largest distance, the points are collinear. If not, they form a triangle instead.

Example 8. Show that A (0, 0), B (2, 2) and C (4, 4) are collinear.

AB = sqrt(4 + 4) = sqrt(8) = 2 sqrt(2). BC = sqrt(4 + 4) = 2 sqrt(2). AC = sqrt(16 + 16) = sqrt(32) = 4 sqrt(2). AB + BC = 2 sqrt(2) + 2 sqrt(2) = 4 sqrt(2) = AC. So the points **are collinear**, with B in the middle - in fact B is also the midpoint of AC.

Example 9. Are A (0, 0), B (3, 0) and C (0, 4) collinear?

AB = 3, AC = 4, BC = sqrt(9 + 16) = 5. Check: 3 + 4 = 7, which is not 5; 3 + 5 is not 4; 4 + 5 is not 3. No two distances add to the third, so the points are **not collinear** - they are the corners of a triangle.

Work in surds, not decimals. 2 sqrt(2) + 2 sqrt(2) = 4 sqrt(2) is exact. Rounded decimals like 2.83 + 2.83 = 5.66 against 5.66 can look equal (or unequal) by accident. Keep the square-root form until the final comparison.

Midpoint link: If B is the midpoint of AC, then A, B and C are automatically collinear and AB = BC. The midpoint formula is therefore a quick collinearity check whenever the middle point is given.

6. Kinds of Triangle by Their Sides

Given the three vertices of a triangle, calculate the three side lengths with the distance formula, then **compare** them. The comparison names the triangle - no drawing needed.

Kind	Test on the three side lengths
Equilateral	All three sides equal: AB = BC = CA.
Isosceles	Exactly two sides equal.
Scalene	All three sides different.
Right-angled	Square of the longest side = sum of the squares of the other two (Pythagoras check). A triangle can be both isosceles and right-angled.

Example 10 - Isosceles. A (1, 2), B (4, 6), C (7, 2).

AB = sqrt(9 + 16) = 5. BC = sqrt(9 + 16) = 5. CA = 6 (same y, gap in x). Two sides equal, so triangle ABC is **isosceles**.

Example 11 - Right-angled. A (0, 0), B (4, 0), C (0, 3).

AB = 4, AC = 3, BC = sqrt(16 + 9) = 5. Check: $3^2 + 4^2 = 9 + 16 = 25 = 5^2$. The Pythagoras check holds, so the triangle is **right-angled at A**, the vertex opposite the longest side BC.

Example 12 - Scalene and equilateral.

A (0, 0), B (5, 0), C (2, 3): AB = 5, AC = sqrt(4 + 9) = sqrt(13), BC = sqrt(9 + 9) = sqrt(18). All three different - **scalene**.

A (0, 0), B (2, 0), C (1, sqrt(3)): AB = 2, AC = sqrt(1 + 3) = 2, BC = sqrt(1 + 3) = 2. All three equal - **equilateral**. Equilateral examples usually need a square root in one coordinate; isosceles and right-angled examples with whole numbers are far commoner in the board paper.

7. Quadrilaterals from Coordinates

Four vertices A, B, C, D (taken in order) make a quadrilateral. Three tools identify it: the **four side lengths**, the **two diagonals**, and the **midpoints of the diagonals**.

Shape	Tests that prove it
Parallelogram	Opposite sides equal (AB = CD and BC = DA), or the diagonals bisect each other - both diagonals have the same midpoint.
Rectangle	A parallelogram whose diagonals are equal in length.
Rhombus	All four sides equal, with diagonals of different lengths.
Square	All four sides equal and the diagonals equal - a rhombus that is also a rectangle.

Example 13 - Parallelogram. A (0, 0), B (3, 1), C (6, 4), D (3, 3).

Midpoint of diagonal AC: ((0 + 6) / 2, (0 + 4) / 2) = (3, 2). Midpoint of diagonal BD: ((3 + 3) / 2, (1 + 3) / 2) = (3, 2). The diagonals share one midpoint, so they bisect each other and ABCD is a **parallelogram**. (Sides check: AB = CD = sqrt(10) and BC = DA = sqrt(18).)

Example 14 - Rectangle and square.

A (0, 0), B (4, 0), C (4, 3), D (0, 3): sides 4, 3, 4, 3 and both diagonals sqrt(16 + 9) = 5. Opposite sides equal with equal diagonals - a **rectangle**.

A (0, 0), B (2, 0), C (2, 2), D (0, 2): all sides 2 and both diagonals sqrt(8) = 2 sqrt(2). All sides equal with equal diagonals - a **square**.

Order matters: Always take the vertices in order around the shape - A to B to C to D and back to A. Joining them in the wrong order pairs up the wrong sides and diagonals, and every test fails.

8. Worked Mixed Examples

Example 15. Find the value of y if the distance between P (2, -3) and Q (4, y) is 2 sqrt(5).

$PQ^2 = (4 - 2)^2 + (y + 3)^2 = 4 + (y + 3)^2$. Set $PQ^2 = (2\ sqrt(5))^2 = 20$. So $(y + 3)^2 = 16$, giving y + 3 = 4 or y + 3 = -4. Hence **y = 1 or y = -7** - two answers, because Q can sit above or below P at the same distance.

Example 16. Show that A (-1, 1), B (3, 4) and C (5, -1) do not form an equilateral triangle.

AB = sqrt(16 + 9) = 5. BC = sqrt(4 + 25) = sqrt(29). CA = sqrt(36 + 4) = sqrt(40). The three lengths 5, sqrt(29) and sqrt(40) are all different, so the triangle is scalene, **not equilateral**. One unequal pair is already enough to disprove "equilateral" - the full working simply makes the answer certain.

Example 17. The points A (2, 3), B (6, 3) and C (4, 7) form a triangle. Find its kind and its perimeter.

AB = 4 (same y). AC = sqrt(4 + 16) = sqrt(20) = 2 sqrt(5). BC = sqrt(4 + 16) = 2 sqrt(5). Two sides equal, so the triangle is **isosceles**. Perimeter = 4 + 2 sqrt(5) + 2 sqrt(5) = **4 + 4 sqrt(5) units**, about 12.94 units.

A routine for every question in this unit

1. Write the two points with labels: (x1, y1) and (x2, y2).
2. Write the formula first - the examiner awards a mark for it.
3. Substitute carefully, one gap per line, watching negative signs.
4. Simplify the square root where possible, for example sqrt(20) = 2 sqrt(5).
5. Finish with the comparison or statement the question asked for: the kind of shape, "collinear" or "not collinear", with the evidence shown.

9. Formula Summary and Common Mistakes

Idea	Rule
Distance between (x1, y1) and (x2, y2)	sqrt($(x2 - x1)^2 + (y2 - y1)^2$)
Distance of (x, y) from origin	sqrt($x^2 + y^2$)
Midpoint of a segment	((x1 + x2) / 2 , (y1 + y2) / 2) - add and halve.
Missing endpoint	Twice the midpoint coordinate minus the known endpoint coordinate.
Collinear points	Two smaller distances add up to the largest distance.
Triangle by sides	All equal: equilateral. Two equal: isosceles. All different: scalene. Longest^2 = sum of other squares: right-angled.
Parallelogram test	Opposite sides equal, or both diagonals have the same midpoint.
Rectangle and square	Parallelogram with equal diagonals is a rectangle; add all sides equal for a square.

Five mistakes that lose marks

1. Forgetting the square root at the end of the distance formula - sqrt(25) must become 5.
2. Squaring wrongly: $(x2 - x1)^2$ means square the whole gap, and $(-8)^2 = 64$, not -64.
3. Using the midpoint formula when distance is asked, or the other way round - subtract and square for distance, add and halve for midpoint.
4. Subtracting negative coordinates wrongly, such as writing 4 - (-2) = 2 instead of 6.
5. Rounding surds early in a collinearity or shape proof, so exact equalities are hidden.

> **One-line revision:** Distance subtracts and squares under one root, midpoint adds and halves - and every shape in this unit is proved by comparing lengths.

10. Practice - MCQs and Questions

Multiple-choice questions

1. The distance of the point (3, 4) from the origin is: (a) 5 (b) 6 (c) 7 (d) 25
2. The distance between (1, 2) and (4, 6) is: (a) 3 (b) 4 (c) 5 (d) 7
3. The midpoint of (2, 6) and (6, 10) is: (a) (3, 7) (b) (4, 8) (c) (8, 16) (d) (4, 4)
4. Three points are collinear when: (a) all distances are equal (b) the two smaller distances add to the largest (c) their midpoint is the origin (d) they lie in one quadrant
5. A triangle with sides 5, 5 and 6 is: (a) equilateral (b) scalene (c) isosceles (d) right-angled
6. If the diagonals of a quadrilateral have the same midpoint, the quadrilateral is a: (a) triangle (b) parallelogram (c) circle (d) trapezium only

Short and long questions

1. Find the distance between (-1, 2) and (3, 5), and the distance of (5, 12) from the origin.
2. Find the midpoint of the segment joining (-4, 3) and (6, -5). If M (2, 2) is the midpoint of AB and A is (0, 5), find B.
3. Show that (1, 1), (3, 3) and (6, 6) are collinear.
4. Classify the triangle with vertices (0, 0), (6, 0) and (0, 8) by its sides and angles.
5. Show that (0, 0), (2, 1), (4, 3) and (2, 2) are the vertices of a parallelogram.

Answer key

MCQs: 1(a), 2(c), 3(b), 4(b), 5(c), 6(b).

Questions: 1) sqrt(16 + 9) = 5 units; sqrt(25 + 144) = 13 units. 2) Midpoint (1, -1); B = (4, -1). 3) Distances 2 sqrt(2), 3 sqrt(2) and 5 sqrt(2), and 2 sqrt(2) + 3 sqrt(2) = 5 sqrt(2), so collinear. 4) Sides 6, 8 and 10; all different (scalene) and $6^2 + 8^2 = 100 = 10^2$, so also right-angled. 5) Midpoint of both diagonals is (2, 1.5), so the diagonals bisect each other - a parallelogram.

> **Final tip:** In the board paper, write the formula, show each gap on its own line, and end with the statement the question asked for. In this unit the comparison sentence - equal, not equal, adds up - is where the final mark lives.